MODEL ANSWERS

1. Variable renewable energy and storage

a) Part A

i) Onshore Wind Farm 0.32 [1]; Offshore wind farm 0.36 [1]

ii) Difference is because offshore generally has better wind resources and there are less obstacles and bigger turbines which are taller and the resource at height is better than onshore [1]

iii) Onshore wind farm 7 TWh; Offshore wind farm 6.36 TWh [1]

iv) It is impossible to say as offshore is more expensive to build but with higher capacity factors it may be worth it. [2]

b) Part B

i) Solar farm 0.11; Rooftop solar 0.102 [2]

ii) Solar farm is higher because of tracking, cleaning of the PV , less likely to be shaded etc. [1]

iii) Solar PV farm 10.08 TWh; Rooftop solar PV 4.5 TWh [1]

iv) Large solar PV farm here has a higher capacity factor and can gain advantages from economies of scale while rooftop solar cannot so solar farm LCOE should be substantially lower. Rooftop solar may also have permitting issues. [2]

c) Part C The wind farm estimates are only based on one year while the solar is based on five years so the solar PV estimates are likely to be more accurate. [2]

d) Part D

i) 20.26 TWh [2]

ii) Potentially yes as between 8AM to noon and noon to 4PM there may be an excess of wind/solar 2.1GWh and 2.5 GWh on average. [1]

iii) Generation = 7 + 10.07 = 17.07 TWh This is short 3.18 TWh to serve the demand so more generation needed. Answers that provide explanations based on shortfalls in the hours between 8 am and 4 pm, instead of annual shortfalls, will also count as correct. [2]

iv) Solar PV may not be a good idea as it is the main reason renewable curtailment is happening. Storage may be a good idea as there are periods of curtailment but it will not make up for the shortfall in energy. Gas is by far the best option because it can make up for the shortfall. [3]

e) Part E

i) 6.48TWh [2]

ii) Every time period has an excess and overall the excess is 4.37 TWh over the year so yes. [1]

iii) No as there is an enormous excess BUT that still does not mean we can serve the demand reliably. There is a need for more granular data. Explanations mentioning potentially stability issues will also be acceptable. [2]

iv) Solar PV is a very bad idea too much curtailment as it is Storage is possibly the best idea as the curtailed energy can be stored for times that there is not enough Gas may not be needed. If students identify a need for gas, citing potential stability issues, it should be considered as correct. [3]

v) Yes as the excess in Region B 4.37 TWh on a yearly basis is bigger than the shortfall in Region A 4.12 TWh but that is not the whole story as it depends on the correlation between excess and shortfall so it will not solve the problem entirely. [3]

2. Modelling demand for electric vehicles in a capacity expansion problem

a) Part A

i) The supply-demand balance constraint $\sum_g gen_{g,t} + nse_t = Demand_t$. [2]

ii) $\sum_g gen_{g,t} + nse_t = Demand_t + EV_t$. [2]

b) Part B

i) $chEV_t$ EV charging at time t [1]
$x_{\tau,t}$ portion of EV demand at time τ that shifts to t [1]

ii) $\sum_g gen_{g,t} + nse_t = Demand_t + chEV_t \ \forall$ t [0.5]
$\sum_{\tau=t:t+4} x_{t,\tau} = 1 \ \forall t$ [1]
$chEV_t = \sum_{\tau=t-4:t} EV_\tau \cdot x_{\tau,t} \forall t$ [0.5]

c) Part C

i) $\sum_g gen_{g,t} + nse_t = Demand_t + chEV_t$. [1]
$chEV_t + sEV_t = EV_t$ [1]
$sEV_t \leq 0.1 \cdot EV_t$ [1]
new decision variable sEV_t:EV demand unmet when price exceeds 1,000$/MWh. [1]

ii) $\sum_t sEV_t \cdot 1000$ [2]

d) Part D

i) True [2]

ii) The feasible region of the problem expands with bidirectional plug-in points. Hence, the generation costs should be higher when plug-in points are uni-directional. [2]
The bi-directional plug-in points cost more than uni-directional points. [2]

iii) 7,000/37=£189 per bidirectional point [1]

iv) 4,000/37=£108 per unidirectional point [1]

e) Part E

i) (41,000-37000)/(0.5*37)=£216.22 per used bidirectional point [2]

ii) Decision node – square [1]

Correct branches after decision node [2]
Chance node – round [1]
Branches after chance node [1], probabilities [1]

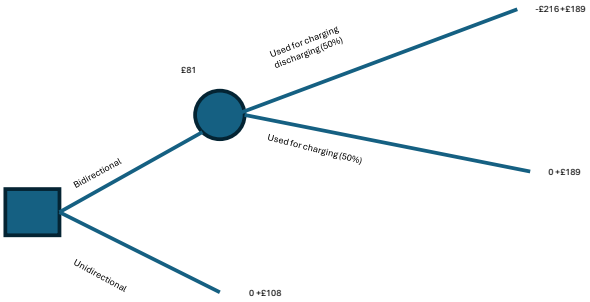


Figure 2.1 Decision tree

iii) With perfect information, the utility will buy: 18.5 million bidirectional chargers (18.5*189) 18.5 million unidirectional chargers (18.5*108) 5.5 billion in total [2]
Comparing this to 7 billion spent when all chargers are bidirectional, the utility would pay up to 1.5 billion for this information. [1]
EVPI= RP-PF [1]

3. Investment plans under uncertainty

a) Part A

i) $EUE = \sum_y prob_y \cdot UE_y = \sum_y UE_y / 34$ [2]
Portfolio 1: 1,786 MWh; Portfolio 2: 2,225 MWh; Portfolio 3: 2561 MWh; Portfolio 4: 2,231 MWh [0.5 point per portfolio]

ii) $\min z s.t F(z) \geq 0.95$
0.95*34=32.3. This means that the value at risk for unserved energy will be given by the second most severe in terms of unserved energy weather year. [2]
Portfolio 1: 2,440 MWh; Portfolio 2: 13,270 MWh; Portfolio 3: 280 MWh; Portfolio 4: 0 MWh [0.5 per portfolio]

iii) $CVAR = ((0.95 - F(VaR)) \cdot VaR + (1 - F(VaR)) \cdot Worst)/0.05$ [2]
Portfolio 1: 34,558 MWh; Portfolio 2: 39,294 MWh; Portfolio 3: 51,180 MWh; Portfolio 4: 44,629 MWh [0.5 per portfolio]

iv) Trade-off curve [2]

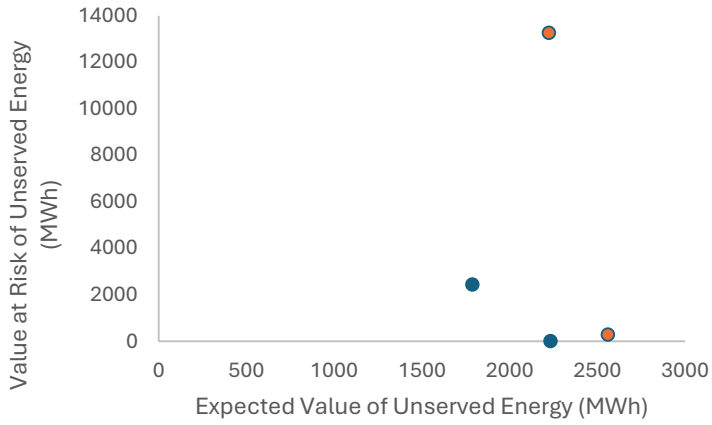


Figure 3.1 Graph with all four portfolios

Dominated alternatives: portfolio 2 & 3 [2]
Non-dominated alternatives: portfolio 1 & 4 [2]

v) See Figure below. Note that I have assumed that I use 0.0001 in the place of 0 because it is more convenient to have a logarithmic scale for x axis. In practice, the settings of the optimisation model will allow you to know if the zero corresponds to 10^{-4} or other small number.

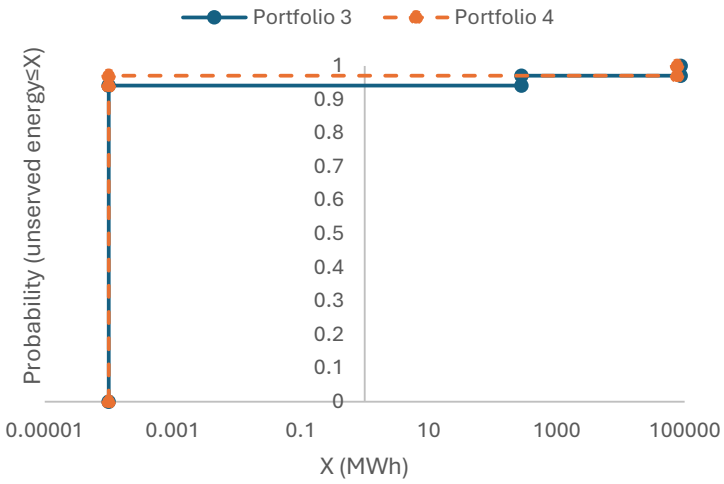


Figure 3.2 Cumulative risk profiles for portfolios 3 and 4

vi) The cumulative risk profiles do not cross. Portfolio 4 stochastically dominates portfolio 3. [2]

b) Part B

i) Constraint: $\sum_n PTDF_{l,n} \cdot NI_n \leq LIMIT_l$ [1]
Parameters: $PTDF_{l,n}$, $LIMIT_l$ [1]

ii) The constraint for that line l should not be included [1]
In the B-theta formulation, no parameter change. In the PTDF-based, the PTDF needs to be recalculated. [1]

iii) The shadow price would be 0 or variable cost for renewables [1]
To formally justify the answer, you can use the stationarity condition.
The $R_{r,h}$ appears in two constraints and in the objective function.
In the objective function, if a variable cost (VC) for renewable production is considered, then $(df_0)/(dR_{r,h}) = VC$.
The shadow price $\mu_{r,h}$ of $R_{r,h} \leq CAP_r \cdot CF_{r,h}$ is 0 because the constraint is not binding.
The shadow price of supply-demand balance at that node is λ_h^E.
The stationarity condition for the renewable production is:
$VC + \mu_{r,h} - \lambda_h^E = 0$ [2]